



Statistical Investigation of Chronic Kidney Disease Risk Factors and Prediction Using Python

Title : Statistical Investigation of Chronic Disease Risk Factors and Prediction

Course Code : STA281.10

Course Title : Statistical Methods and Probability

Department : CSE

FacultyName : Md. Shohel Babu

Assignment Submission Date : 09 June 2026

Submitted By:

GROUP NAME : Tehreek-e-Muhandis

SL	Student Name	ID
1	MD. RAYHAN KAMAL	2024100000250
2	SAIFUL SIDDIKY LABIB	2024100000469
3	ISTIAK AHAMED	2024100000273
4	ABDUL AHAD EMON	2024100000490
5	MD SHAH NAOHAJ KHAN	2024100000487
6	ABDUL AHAD	2024100000140

Overview:

Page no :	Topic:
1	COVER PAGE
2	OVERVIEW
3	INTRODUCTION
4	OBJECTIVES
5	REVIEW
6	DATASET
6-7	VARIABLES DESCRIPTION
8	DATASET SUMMARY
9	METHODOLOGY
10-15	GRAPHS AND VISUALIZATIONS
16	BAYES' THEOREM
16	DISCUSSION
17	LIMITATIONS
17	MISSING DATA ISSUES
18-32	13.FINDINGS AND HEALTH CARE INTERPRETATIONS

1.INTRODUCTION:

Chronic Kidney Disease (CKD) is a progressive medical condition in which the kidneys gradually lose their ability to filter waste products and excess fluids from the blood. As kidney function declines, harmful substances can accumulate in the body, leading to serious health complications such as hypertension, anemia, cardiovascular diseases, and kidney failure. CKD is recognized as a significant public health issue worldwide due to its increasing prevalence and impact on quality of life.

Early detection of CKD is essential for preventing disease progression and reducing the risk of severe complications. However, the disease often develops without noticeable symptoms in its early stages, making timely diagnosis challenging. Several factors, including age, blood pressure, blood glucose levels, hemoglobin levels, and other clinical indicators, are known to influence the risk of developing CKD. Understanding the relationship between these factors and the disease can help healthcare professionals make more informed decisions regarding diagnosis and treatment.

The rapid growth of healthcare data has created opportunities to apply statistical analysis and machine learning techniques in medical research. Statistical methods can be used to identify significant risk factors and examine relationships between variables, while machine learning models can analyze complex patterns within data to predict disease outcomes with high accuracy. These approaches provide valuable insights that can support early diagnosis and improve patient care.

This project focuses on the statistical investigation of Chronic Kidney Disease risk factors and the development of predictive models using Python. The study involves data preprocessing, exploratory data analysis, statistical testing, and machine learning model development. By analyzing patient data and evaluating different predictive techniques, the project aims to identify the most important risk factors associated with CKD and assess the effectiveness of prediction models. The findings of this study may contribute to a better understanding of CKD and demonstrate the role of data-driven methods in supporting healthcare decision-making.

2.OBJECTIVES:

OBJECTIVE NO	CRITERIA
1	Identify significant risk factors associated with Chronic Kidney Disease.
2	Perform exploratory data analysis using descriptive statistics.
3	Apply probability concepts and Bayes' theorem in healthcare analytics.
4	Analyze probability distributions of medical indicators.
5	Estimate population parameters using confidence intervals.
6	Conduct hypothesis testing to evaluate statistical significance.
7	Examine relationships among variables through correlation analysis.
8	Build a Logistic Regression model for CKD prediction.
9	Interpret findings from both statistical and medical perspectives.

3.REVIEW:

1.Previous Studies on CKD Prediction:

Chronic Kidney Disease (CKD) has been widely studied due to its growing impact on global health. Researchers have used clinical and demographic data to identify factors associated with CKD and to develop predictive models for early diagnosis. Previous studies have shown that variables such as age, blood pressure, diabetes, hemoglobin levels, serum creatinine, and albumin are strongly related to the occurrence of CKD. Early prediction of the disease has been recognized as an effective approach for reducing complications and improving patient outcomes

2.Statistical Techniques Used in Related Research:

Various statistical methods have been applied to analyze CKD datasets and identify significant risk factors. Descriptive statistics are commonly used to summarize patient characteristics and understand data distributions. Correlation analysis helps determine relationships between clinical variables, while hypothesis testing, including t-tests and chi-square tests, is used to assess the significance of these relationships. Logistic regression has also been widely employed to model the probability of CKD occurrence and evaluate the influence of individual risk factors.

3.Machine Learning Approaches for CKD Diagnosis:

Recent studies have increasingly utilized machine learning techniques for CKD prediction due to their ability to handle complex datasets and achieve high predictive accuracy. Commonly used algorithms include **Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Naïve Bayes**. Among these methods, **Random Forest and SVM** have often demonstrated strong performance in CKD classification tasks. These models assist healthcare professionals by providing reliable predictions and supporting early disease detection. The integration of machine learning with statistical analysis has significantly enhanced the effectiveness of CKD diagnosis and risk assessment.

4.Dataset :

Data Source

→ kidney_disease.csv

5.Variables Description:

Variable	Improved Description
Age	Age of the patient in years at the time of clinical assessment.
BloodPressure	Patient's blood pressure measurement, typically expressed in mmHg, indicating cardiovascular health status.
SpecificGravity	Urine specific gravity, reflecting the concentration of dissolved substances in urine and kidney concentrating ability.
Albumin	Urinary albumin level, indicating the presence and severity of protein leakage through the kidneys.
Sugar	Urinary glucose level, indicating the presence of sugar in urine and potential abnormalities in glucose metabolism.
BloodGlucoseRandom	Random blood glucose concentration measured irrespective of fasting status, used to assess blood sugar control.
BloodUrea	Blood urea concentration, a biochemical marker used

Variable	Improved Description
	to evaluate kidney function and nitrogen waste excretion.
SerumCreatinine	Concentration of creatinine in the blood, commonly used to assess kidney filtration efficiency and renal function.
Sodium	Serum sodium concentration, an essential electrolyte involved in fluid balance, nerve function, and cellular activity.
Potassium	Serum potassium concentration, an important electrolyte required for proper muscle, nerve, and cardiac function.
Hemoglobin	Hemoglobin concentration in blood, indicating the oxygen-carrying capacity of red blood cells and anemia status.
PackedCellVolume	Percentage of blood volume occupied by red blood cells (hematocrit), used to evaluate anemia and hydration status.
WhiteBloodCellCount	Total number of white blood cells in the blood, reflecting immune system activity and possible infection or inflammation.
RedBloodCellCount	Total number of red blood cells in the blood, used to assess oxygen transport capacity and hematological health.
Hypertension	Presence or absence of diagnosed high blood pressure.
DiabetesMellitus	Presence or absence of diabetes mellitus, a chronic metabolic disorder characterized by elevated blood glucose levels.

6.Dataset Summary:

The dataset contains 400 observations and 26 variables, predictor variables and one target variable (Class), which indicates whether a patient is diagnosed with **Chronic Kidney Disease (CKD)** or Not CKD. This makes it a binary classification dataset used for medical diagnosis tasks.

The predictor variables represent a combination of clinical, laboratory, and demographic features, such as blood test results (e.g., blood glucose, serum creatinine, hemoglobin levels), urine analysis indicators (e.g., pus cell, bacteria), and health conditions like hypertension, diabetes mellitus, and anemia. The dataset includes both numerical and categorical variables, requiring appropriate preprocessing before modeling.

Additionally, the dataset contains missing values in several features, which must be addressed through data cleaning techniques such as imputation or removal to ensure reliable model performance. The goal of using this dataset is to build predictive models that can accurately classify CKD cases based on the available medical attributes.

7.Methodology:

1. Data Preprocessing

Before conducting the statistical analysis, the dataset is preprocessed to ensure data quality and consistency. The preprocessing steps include:

- Handling missing values using appropriate imputation techniques.
- Removing duplicate records and correcting inconsistencies.
- Encoding categorical variables into numerical formats suitable for analysis.
- Normalizing or standardizing numerical features where necessary.

2. Descriptive Statistical Analysis

Descriptive statistics are used to summarize the characteristics of the dataset and provide an overview of the patient population. The following measures are calculated:

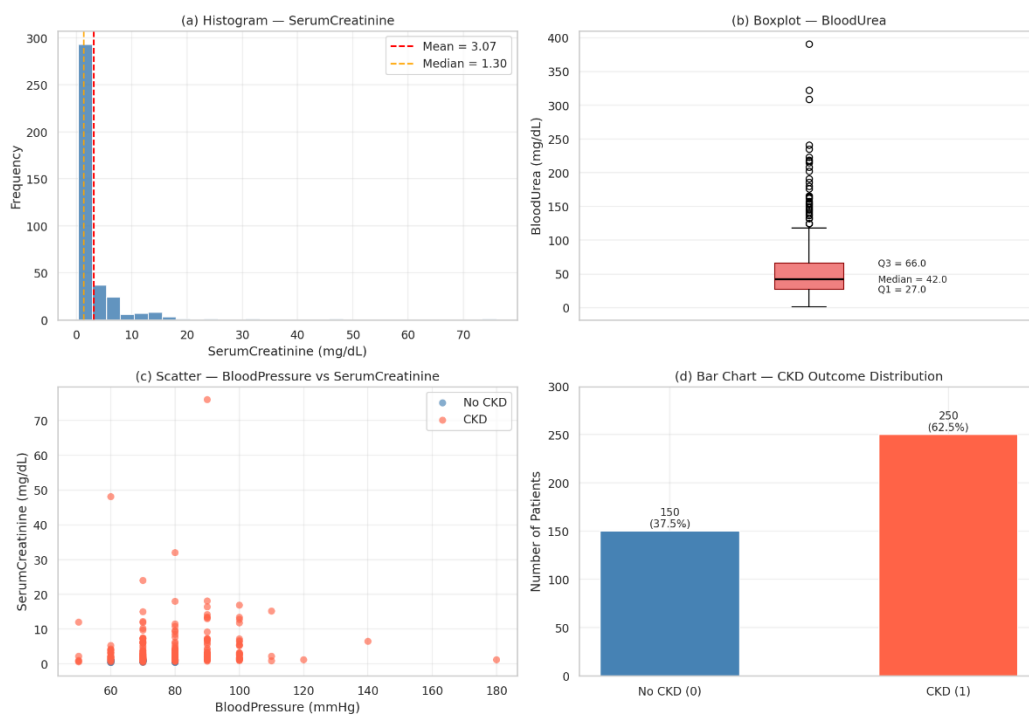
- Mean
- Median
- Mode:
- Standard Deviation
- Variance
- Range
-

These statistics help identify central tendencies, variability, and potential outliers in the data.

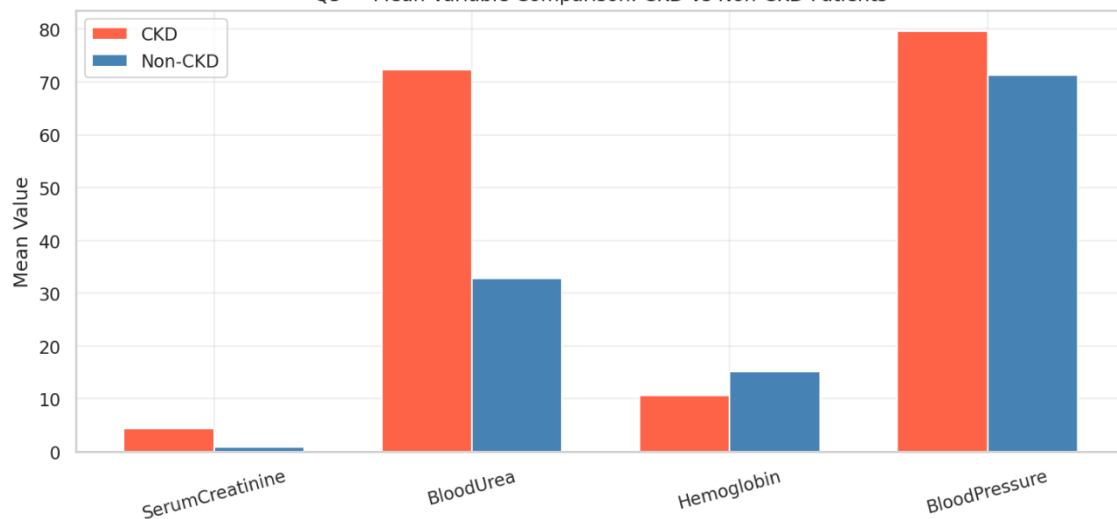
8. Graphs and Visualizations:

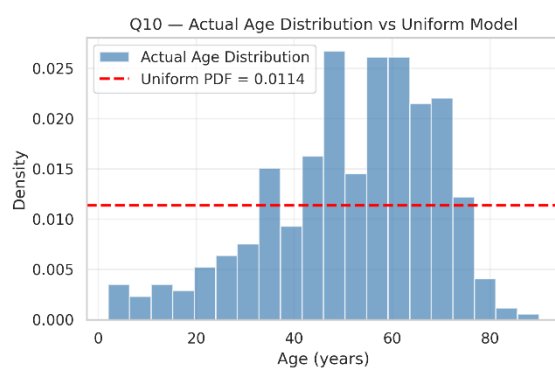
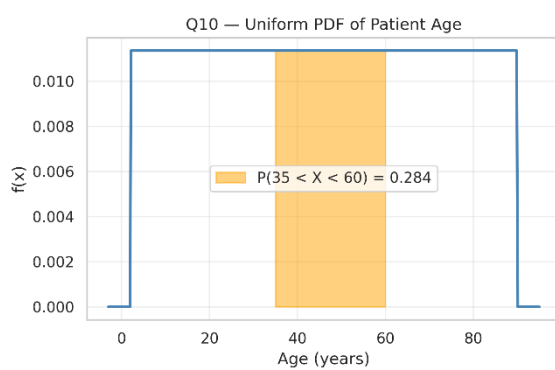
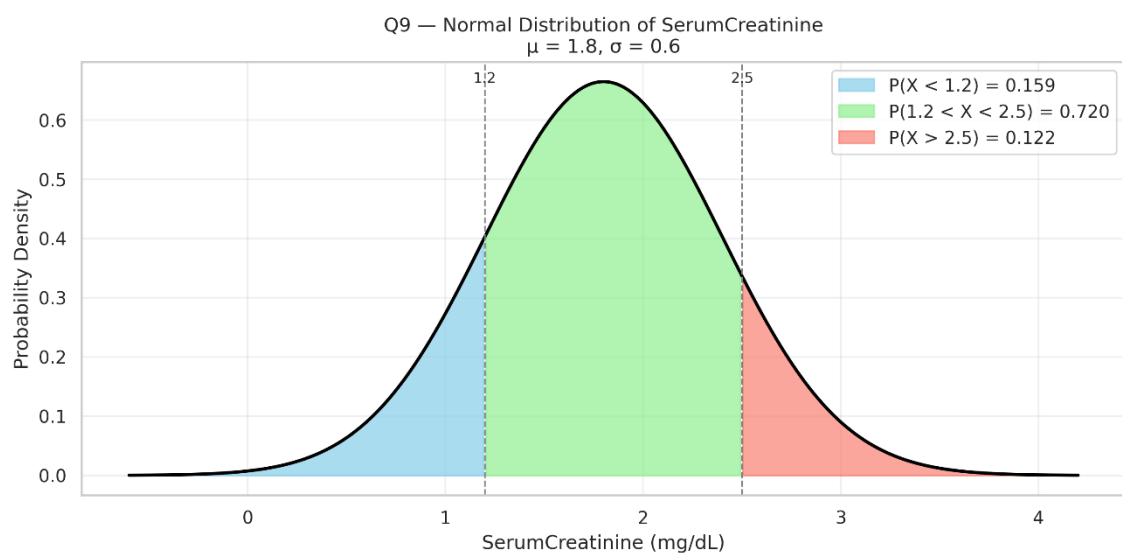
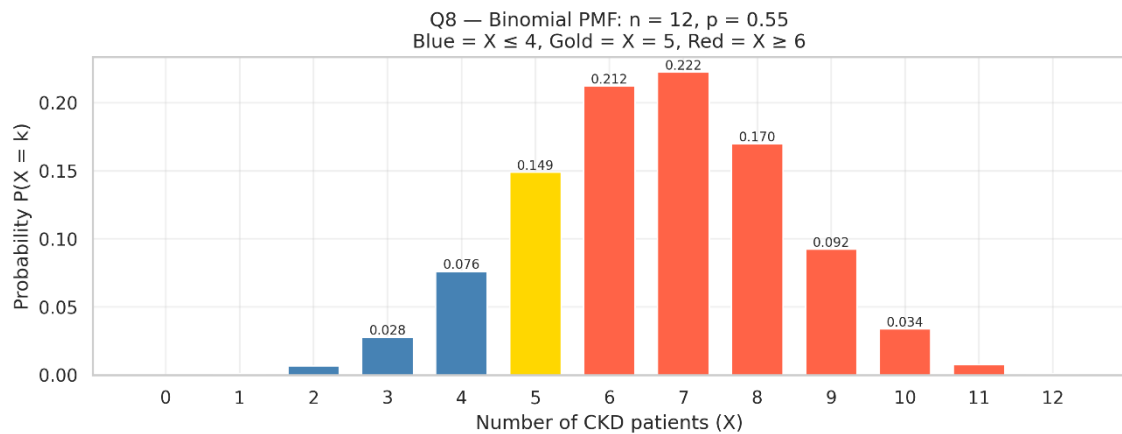
Key Clinical Variables — CKD Patient Dataset (n=400)

Question 4 — CKD Dataset Visualizations

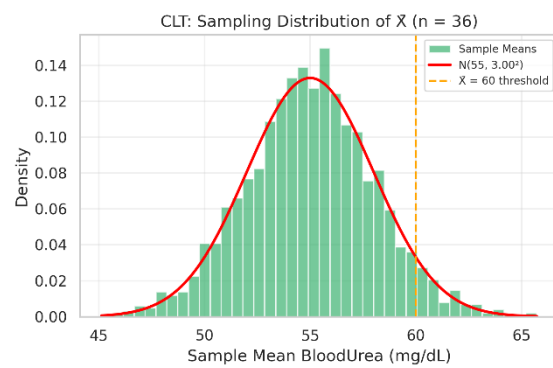
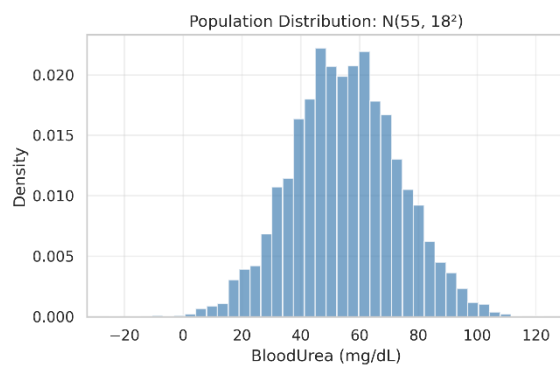
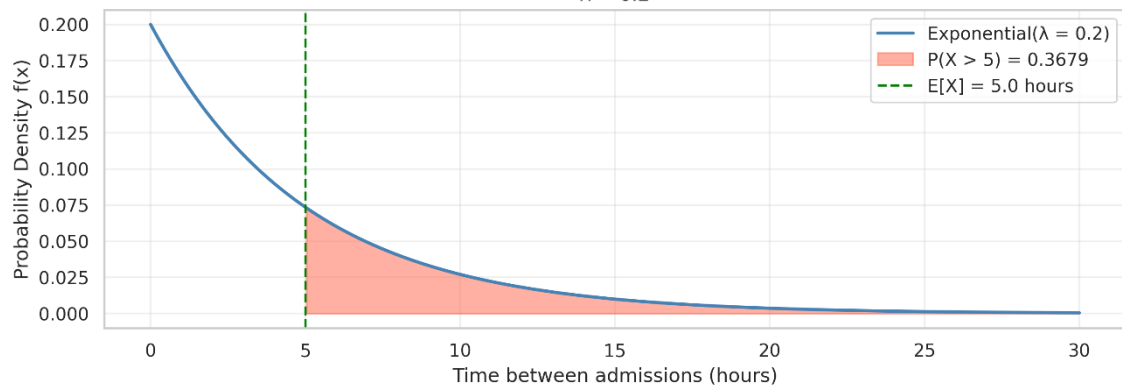


Q5 — Mean Variable Comparison: CKD vs Non-CKD Patients

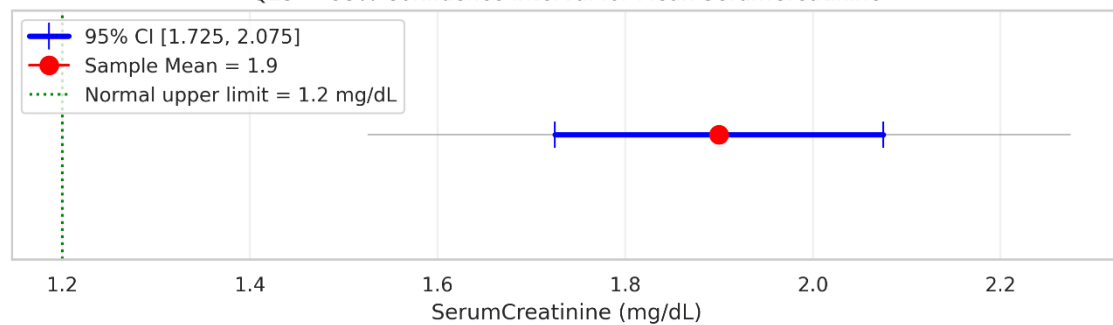


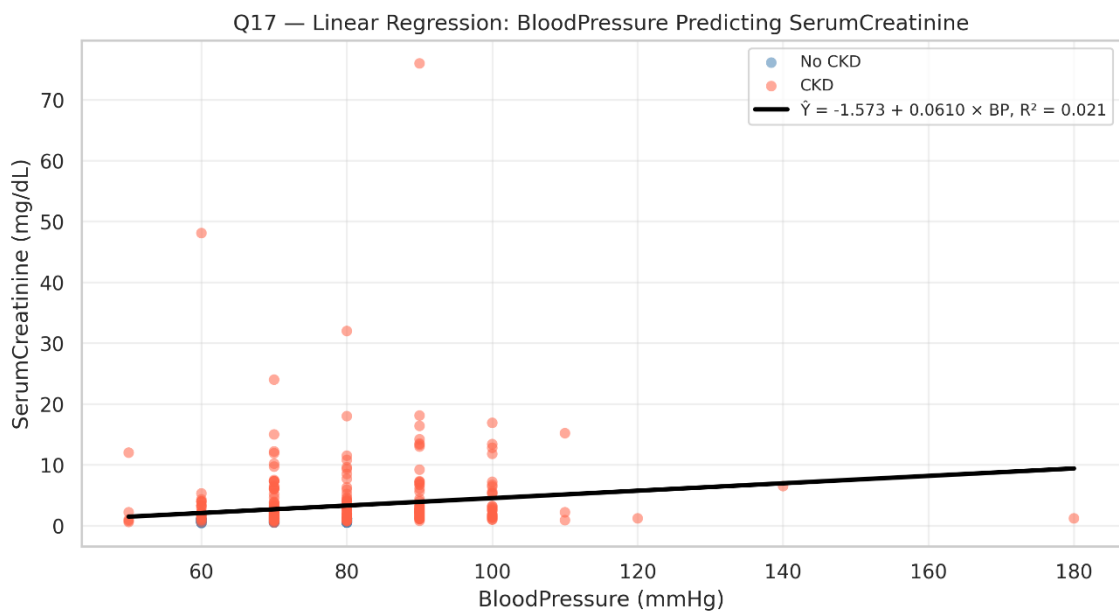
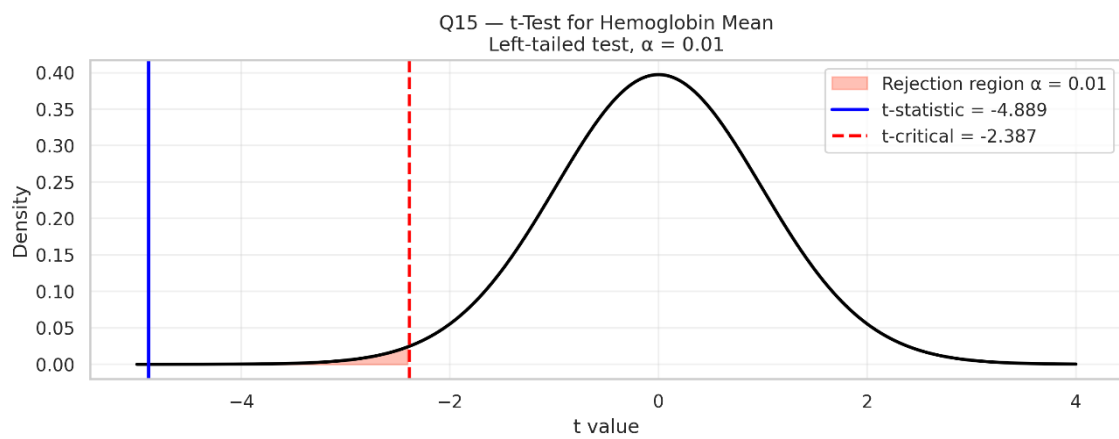
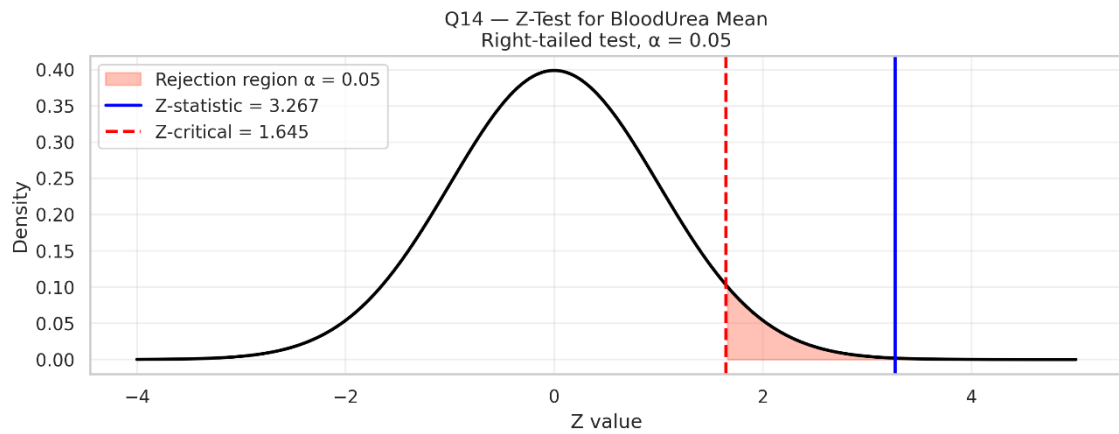


Q11 — Exponential Distribution of CKD Patient Admission Intervals
 $\lambda = 0.2$

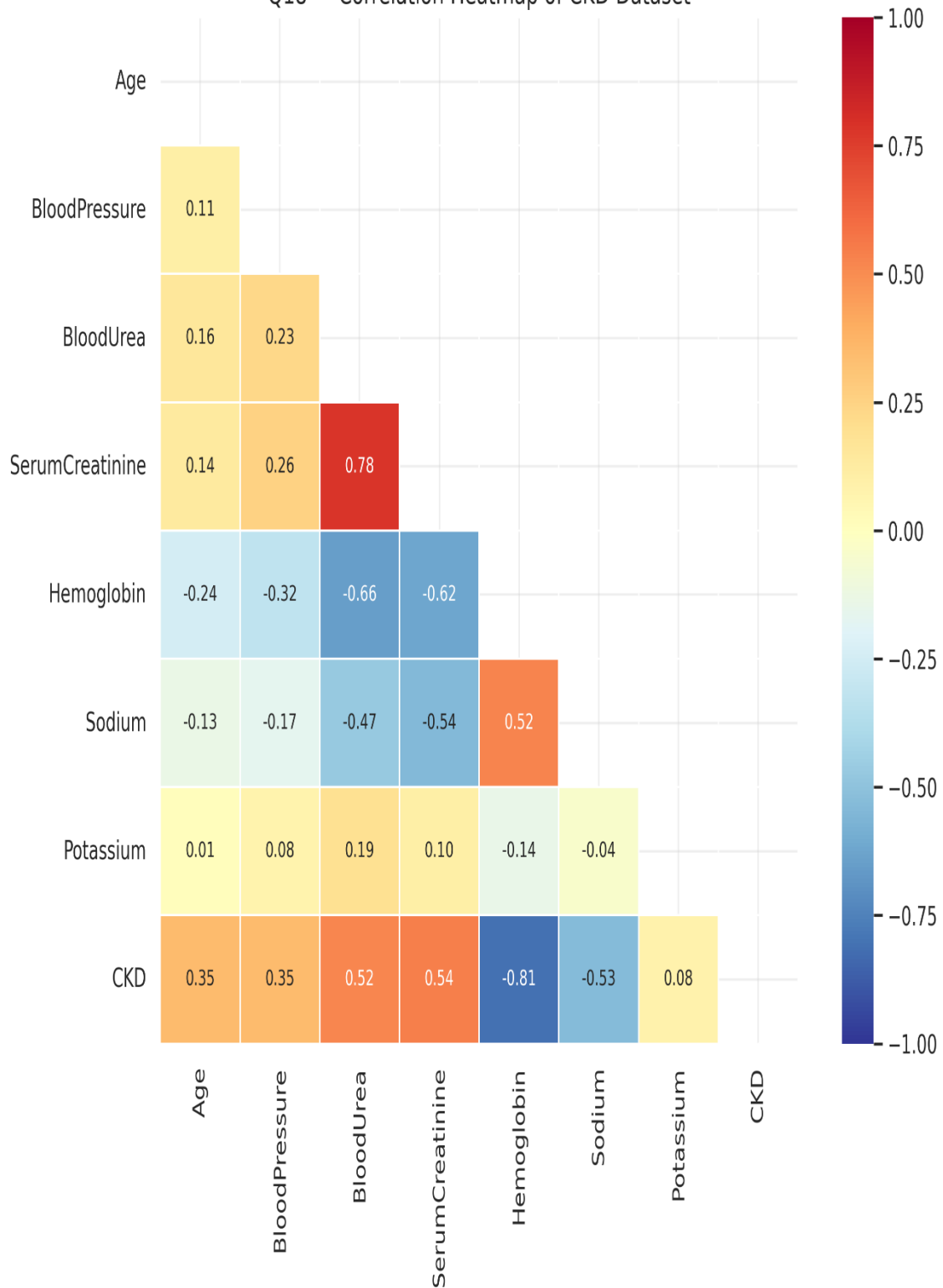


Q13 — 95% Confidence Interval for Mean SerumCreatinine

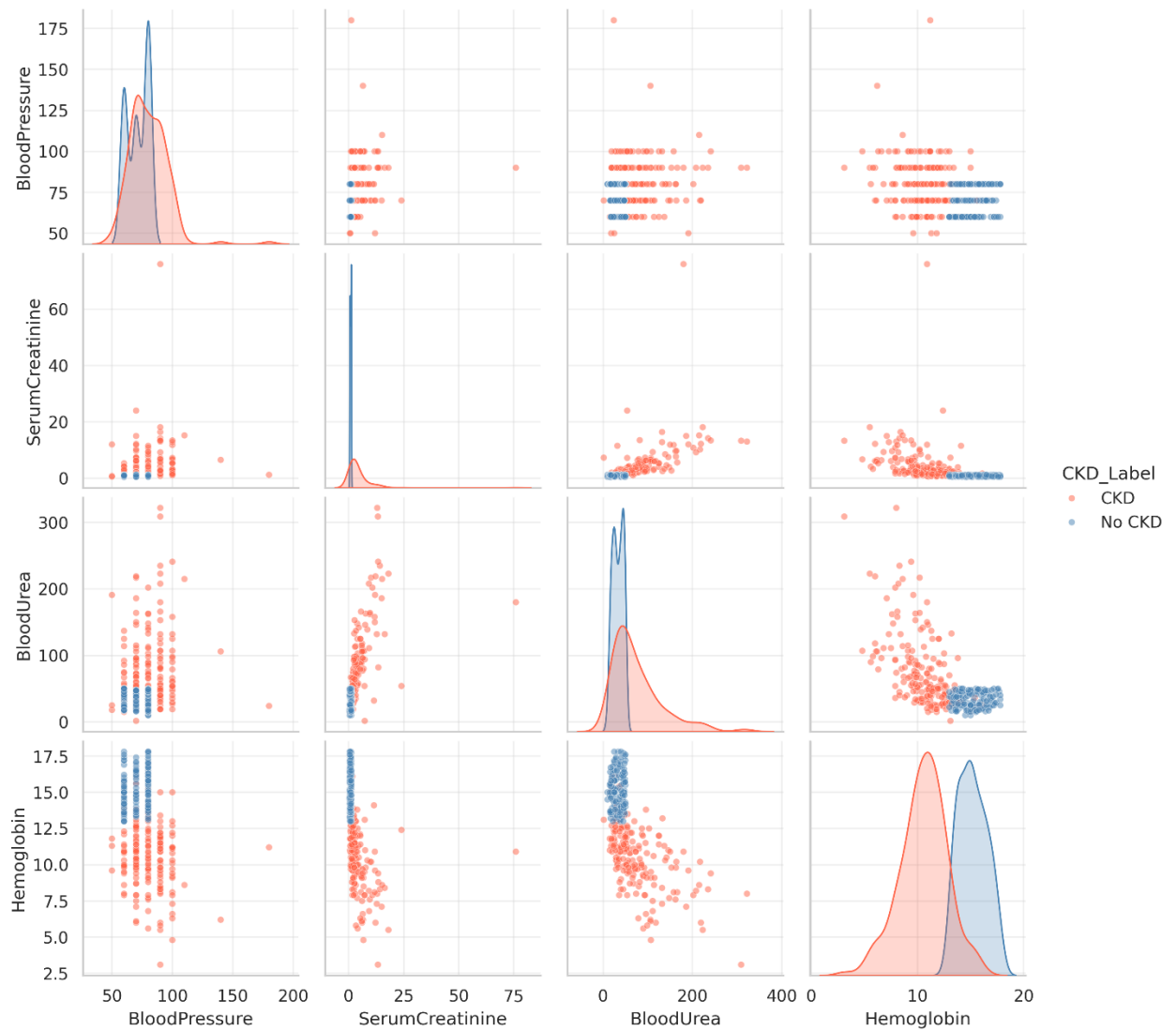




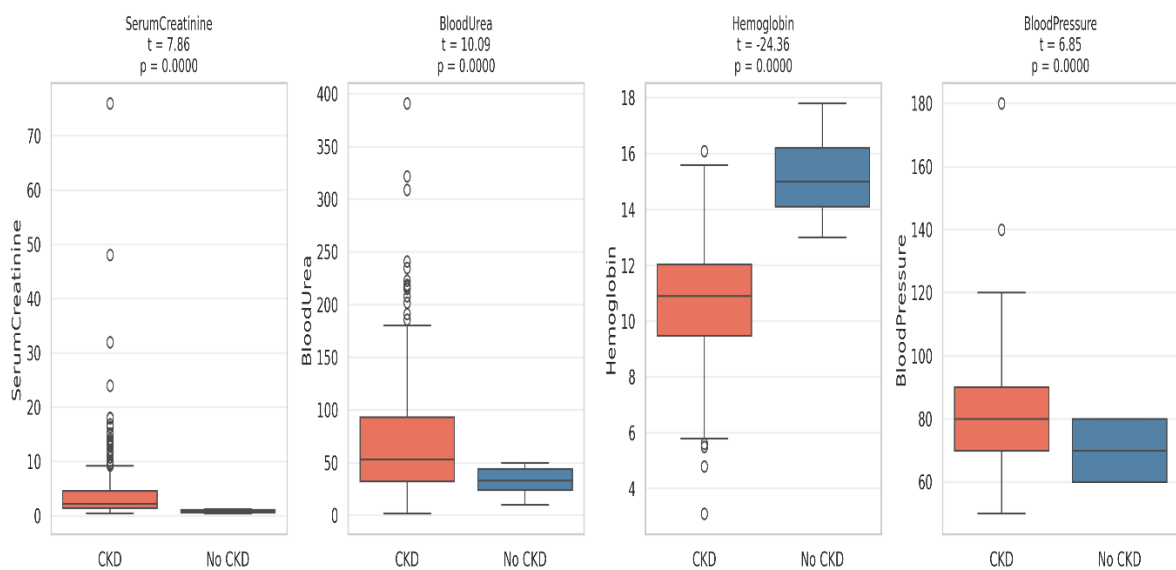
Q18 — Correlation Heatmap of CKD Dataset



Q18 — Scatter Plot Matrix of Key CKD Variables



Q19 — Comparison of CKD and Non-CKD Patients



9. Bayes' Theorem:

Bayes' theorem is used to estimate the probability of a patient having Chronic Kidney Disease (CKD) given the presence of specific risk factors or medical conditions.

$$P(\text{CKD}|\text{RiskFactor}) = \{ P(\text{RiskFactor})P(\text{RiskFactor}|\text{CKD}) \times P(\text{CKD}) \} / P(\text{RiskFactor})$$

Where:

- **P(CKD | Risk Factor)** = Probability of CKD given the risk factor.
- **P(Risk Factor | CKD)** = Probability of observing the risk factor among CKD patients.
- **P(CKD)** = Prior probability of CKD.
- **P(Risk Factor)** = Overall probability of the risk factor in the population.

This theorem helps assess the impact of specific medical conditions on CKD risk.

10. Discussion:

Key findings:

- Serum creatinine and blood urea are strong indicators of CKD.
- Hypertension and diabetes significantly increase CKD risk.
- Hemoglobin levels show negative correlation with CKD occurrence.
- Logistic regression provides effective prediction performance for CKD classification.

11.Limitations:

Dataset size limitations:

When the dataset is too small or not sufficiently diverse, models struggle to learn meaningful patterns.

→**Overfitting risk:** The model may memorize training data instead of learning general patterns.

→**Poor representation:** Small datasets may not include all real-world variations.

→**Low statistical power:** Results may not be reliable or stable.

12.Missing Data Issues:

Since the dataset has some values missing they might slightly skew the results of mean , median , mode etc. However, due to the nature and sample size of the data sets and the number of variables to consider the accuracy of the results should not be that deviated from the actual results .Hence this dataset can still be used to produce acceptable results.

13.FINDINGS AND HEALTH CARE INTERPRETATIONS:

=====

QUESTION 2 – Descriptive Statistics

=====

Descriptive Statistics Summary:

Variable	Mean	Median	Mode	Variance	Std Dev	Min	Max
BloodUrea	57.426	42.00	46.0	2550.554	50.503	1.5	391.0
SerumCreatinine	3.072	1.30	1.2	32.961	5.741	0.4	76.0
Hemoglobin	12.526	12.65	15.0	8.483	2.913	3.1	17.8
BloodPressure	76.469	80.00	80.0	187.242	13.684	50.0	180.0

--- Healthcare Interpretation ---

BloodUrea: Normal range 7-20 mg/dL; elevated mean indicates kidney filtration issues.

SerumCreatinine: Normal 0.6-1.2 mg/dL; high mean signals reduced kidney function.

Hemoglobin: Normal 12-17 g/dL; low mean suggests anemia – common in CKD patients.

BloodPressure: Normal <80 mmHg diastolic; high spread suggests hypertensive patients.

=====

QUESTION 3 – Missing Values & Outlier Detection

=====

Missing Value Analysis:

SerumCreatinine	: 17 missing (4.25%)
BloodUrea	: 19 missing (4.75%)
Hemoglobin	: 52 missing (13.0%)
Potassium	: 88 missing (22.0%)

Outlier Detection using IQR Method:

	Variable	Q1	Q3	IQR	Lower Fence	Upper Fence	Outlier Count
0	SerumCreatinine	0.9	2.8	1.9	-1.95	5.65	51
1	BloodUrea	27.0	66.0	39.0	-31.50	124.50	38
2	Hemoglobin	10.3	15.0	4.7	3.25	22.05	1
3	Potassium	3.8	4.9	1.1	2.15	6.55	4

--- Healthcare Interpretation ---

SerumCreatinine and BloodUrea may contain high outliers, which can indicate patients with severe kidney dysfunction or abnormal laboratory records.

Missing values in healthcare variables may occur due to incomplete laboratory tests, unavailable measurements, or inconsistent patient records.

These missing and abnormal values can affect CKD prediction accuracy because machine learning or statistical models may become biased if incomplete records are not handled properly.

```
=====
QUESTION 4 – Visualizations
=====
```

Data available for visualizations:

SerumCreatinine valid values: 383

BloodUrea valid values: 381

BloodPressure valid values: 388

CKD value counts:

CKD

1 250

0 150

Name: count, dtype: int64

Valid rows for scatter plot: 373

--- Healthcare Interpretation ---

- (a) The SerumCreatinine histogram is right-skewed. Most patients have relatively low creatinine values, while a smaller number of patients have very high values, indicating possible severe kidney dysfunction.
- (b) BloodUrea boxplot summary:
 Q1 = 27.00, Median = 42.00, Q3 = 66.00
 IQR = 39.00
 Number of outliers = 38
 A wide spread and high outliers indicate abnormal urea accumulation, which may be associated with impaired kidney filtration.
- (c) The scatter plot compares BloodPressure with SerumCreatinine.
 CKD and non-CKD patients are shown separately to observe whether CKD patients tend to appear at higher blood pressure and creatinine levels.
- (d) The CKD outcome bar chart compares the number of CKD and non-CKD patients.
 It shows the class distribution of the dataset and helps identify whether the dataset is balanced or imbalanced for prediction analysis.

=====

QUESTION 5 – CKD vs Non-CKD Patient Comparison

=====

CKD patients : 250
Non-CKD patients : 150

CKD value counts:
CKD
0 150
1 250
Name: count, dtype: int64

Mean Comparison Table:

	Variable	CKD Mean	Non-CKD Mean	Difference	% Change
0	SerumCreatinine	4.415	0.869	3.546	408.1
1	BloodUrea	72.389	32.799	39.590	120.7
2	Hemoglobin	10.648	15.188	-4.541	-29.9
3	BloodPressure	79.625	71.351	8.274	11.6

--- Healthcare Interpretation ---

CKD patients show higher average SerumCreatinine and BloodUrea levels than non-CKD patients, which indicates reduced kidney filtration ability.

CKD patients also show lower average Hemoglobin levels, which may reflect anemia caused by reduced erythropoietin production in damaged kidneys.

Higher average BloodPressure among CKD patients supports the relationship between hypertension and kidney disease, where high blood pressure can be both a risk factor and a consequence of CKD.

```
=====
QUESTION 6 – Conditional Probability
=====
```

Total valid patients: 383

$$P(K) = 0.6214$$

$$P(C) = 0.4413$$

$$P(K \cap C) = 0.4413$$

$$P(K | C) = 1.0000$$

--- Healthcare Interpretation ---

The probability of CKD among patients with SerumCreatinine greater than 1.5 mg/dL is 100.00%.

This indicates that elevated serum creatinine is strongly associated with CKD risk.

```
=====
QUESTION 7 – Bayes Theorem
=====
```

Threshold: SerumCreatinine > 1.8 mg/dL

--- Prior Probabilities ---

$$P(\text{CKD}) = 0.6214$$

$$P(\text{No CKD}) = 0.3786$$

--- Likelihoods ---

$$P(\text{SerumCreatinine} > 1.8 | \text{CKD}) = 0.6092$$

$$P(\text{SerumCreatinine} > 1.8 | \text{No CKD}) = 0.0000$$

--- Evidence ---

$$P(\text{SerumCreatinine} > 1.8) = 0.3786$$

--- Posterior Probability ---

$$P(\text{CKD} | \text{SerumCreatinine} > 1.8) = 1.0000$$

Bayes Calculation:

$$\text{Numerator} = 0.6092 \times 0.6214$$

$$\text{Denominator} = 0.3786$$

$$\text{Posterior} = 1.0000$$

--- Healthcare Interpretation ---

A patient with SerumCreatinine greater than 1.8 mg/dL has a 100.0% probability of having CKD.

The baseline CKD probability in the dataset is 62.1%.

This indicates that elevated SerumCreatinine is strongly associated with Chronic Kidney Disease and can be used as an important clinical screening indicator.

=====

QUESTION 8 – Binomial Distribution

=====

Parameters: $n = 12$ patients, $p = 0.55$

$X \sim \text{Binomial}(n = 12, p = 0.55)$

--- Distribution Summary ---

Mean = 6.600

Variance = 2.970

Std Dev = 1.723

--- Results ---

$P(X = 5) = 0.148945$

$P(X \geq 6) = 0.739315$

$P(X \leq 4) = 0.111740$

number of CKD patients (X)

--- Healthcare Interpretation ---

In a group of 12 randomly selected patients with CKD probability 55%,
the probability that exactly 5 patients have CKD is 14.89%.

The probability that at least 6 patients have CKD is 73.93%.

The probability that at most 4 patients have CKD is 11.17%.

This helps estimate expected CKD patient load for hospital planning and resource allocation.

=====

QUESTION 9 – Normal Distribution (SerumCreatinine)

=====

$X \sim N(\mu = 1.8, \sigma = 0.6)$

$P(X < 1.2) = 0.1587 \rightarrow 15.87\%$

$P(1.2 < X < 2.5) = 0.7197 \rightarrow 71.97\%$

$P(X > 2.5) = 0.1217 \rightarrow 12.17\%$

Verification (sum) = 1.0000

--- Healthcare Interpretation ---

$P(X < 1.2) = 15.87\%$ represents patients likely to have normal kidney function.

$P(1.2 < X < 2.5) = 71.97\%$ represents patients with moderate risk who may require monitoring.

$P(X > 2.5) = 12.17\%$ represents patients at high risk of severe kidney dysfunction.

The normal distribution helps estimate the proportion of patients falling into different clinical risk categories based on SerumCreatinine levels.

=====

QUESTION 10 – Uniform Distribution (Patient Age)

=====

Age range from dataset: $a = 2$ years, $b = 90$ years

$X \sim \text{Uniform}(a = 2, b = 90)$

PDF $f(x) = 1/(b-a) = 1/(90-2) = 0.011364$, for $2 \leq x \leq 90$

$P(35 < X < 60) = (60-35)/(90-2) = 0.2841 = 28.41\%$

Mean age $\mu = (a+b)/2 = 46.00$ years

Variance $\sigma^2 = (b-a)^2/12 = 645.3333$

Std Dev $\sigma = \sqrt{\sigma^2} = 25.4034$

--- Healthcare Interpretation ---

Under the uniform distribution assumption, approximately 28.41% of patients are estimated to be between 35 and 60 years old.

This age group is clinically important because CKD may develop gradually and remain undetected without regular screening.

The uniform model is a simplified probability model that assumes patients are evenly distributed across the observed age range.

=====

QUESTION 11 – Exponential Distribution (Patient Admission Intervals)

=====

Parameters: $\lambda = 0.2$ per hour

Scale = $1/\lambda = 5.00$ hours

$X \sim \text{Exponential}(\lambda = 0.2)$

PDF: $f(x) = \lambda e^{-\lambda x} = 0.2e^{-0.2x}$, for $x \geq 0$

$P(X > 5) = e^{-\lambda \times 5} = e^{-1.00} = 0.3679$

$E[X] = 1/\lambda = 5.00$ hours

$\text{Var}[X] = 1/\lambda^2 = 25.00$ hours²

--- Healthcare Interpretation ---

The expected waiting time between consecutive CKD patient admissions is 5.00 hours.

The probability that the next CKD patient arrives after more than 5 hours is 36.79%.

This result can help hospitals estimate patient flow, plan staffing, and allocate nephrology-related resources more effectively.

=====

QUESTION 12 – Central Limit Theorem (BloodUrea)

=====

Population: $\mu = 55$ mg/dL, $\sigma = 18$ mg/dL

Sample size: $n = 36$

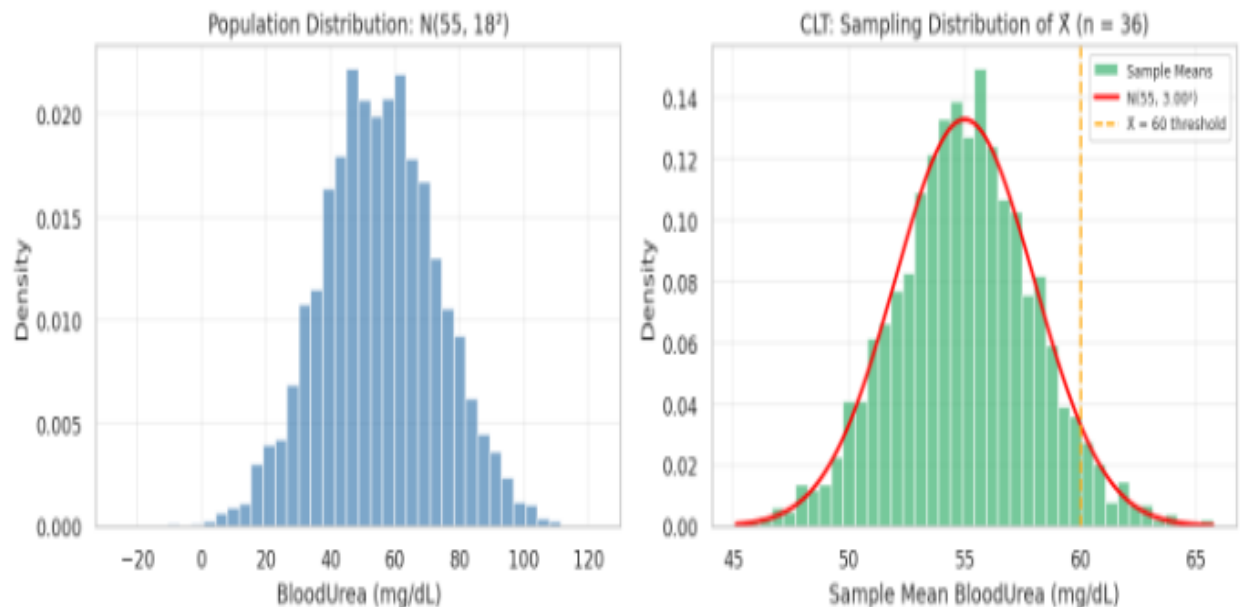
By CLT: $\bar{X} \sim N(\mu, \sigma^2/n) = N(55, 18^2/36)$

Mean of sampling distribution: $\mu_{\bar{X}} = 55$

Standard Error: $SE = \sigma/\sqrt{n} = 18/\sqrt{36} = 3.0000$

Z-score for $\bar{X} = 60$: $Z = (60 - 55) / 3.0000 = 1.6667$

$P(\bar{X} > 60) = P(Z > 1.6667) = 0.0478 = 4.78\%$



--- Healthcare Interpretation ---

The Central Limit Theorem allows the sampling distribution of the sample mean BloodUrea level to be approximated by a normal distribution.

There is a 4.78% probability that a sample of 36 patients will have a mean BloodUrea level greater than 60 mg/dL.

This helps healthcare researchers evaluate whether observed average BloodUrea levels are unusually high and may indicate kidney dysfunction in the patient group.

=====

QUESTION 13 – 95% Confidence Interval (SerumCreatinine)

=====

Given: $n = 64$, $\bar{X} = 1.9$, $s = 0.7$

Confidence level = 95%

Degrees of freedom: $df = n - 1 = 63$

t-critical value = 1.9983

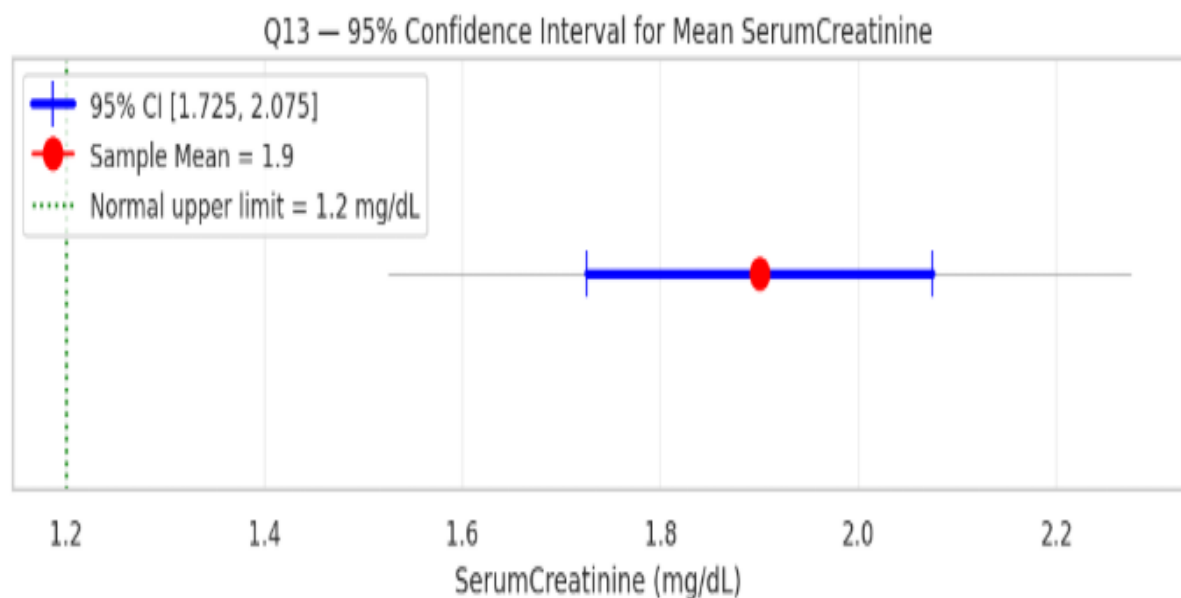
Standard Error: $SE = s/\sqrt{n} = 0.0875$

Margin of Error: $E = t \times SE = 0.1749$

95% Confidence Interval for population mean SerumCreatinine:

[1.7251, 2.0749] mg/dL

1.9000 ± 0.1749



--- Healthcare Interpretation ---

We are 95% confident that the true population mean SerumCreatinine level lies between 1.725 and 2.075 mg/dL.

Since this interval is above the normal upper limit of 1.2 mg/dL, it suggests impaired kidney function among CKD patients.

This confidence interval provides statistical evidence that average SerumCreatinine levels are clinically elevated in the CKD patient group.

=====

QUESTION 14 – Hypothesis Test: Average BloodUrea > 50 mg/dL

=====

$H_0: \mu = 50 \text{ mg/dL}$

$H_1: \mu > 50 \text{ mg/dL}$

Significance level: $\alpha = 0.05$

Test type: One-tailed right-tailed Z-test

Given: $n = 49$, $\bar{X} = 57$, $s = 15$

Standard Error: $SE = s/\sqrt{n} = 2.1429$

Test Statistic: $Z = (\bar{X} - \mu_0) / SE = (57 - 50) / 2.1429 = 3.2667$

Critical Value: $Z_\alpha = 1.6449$

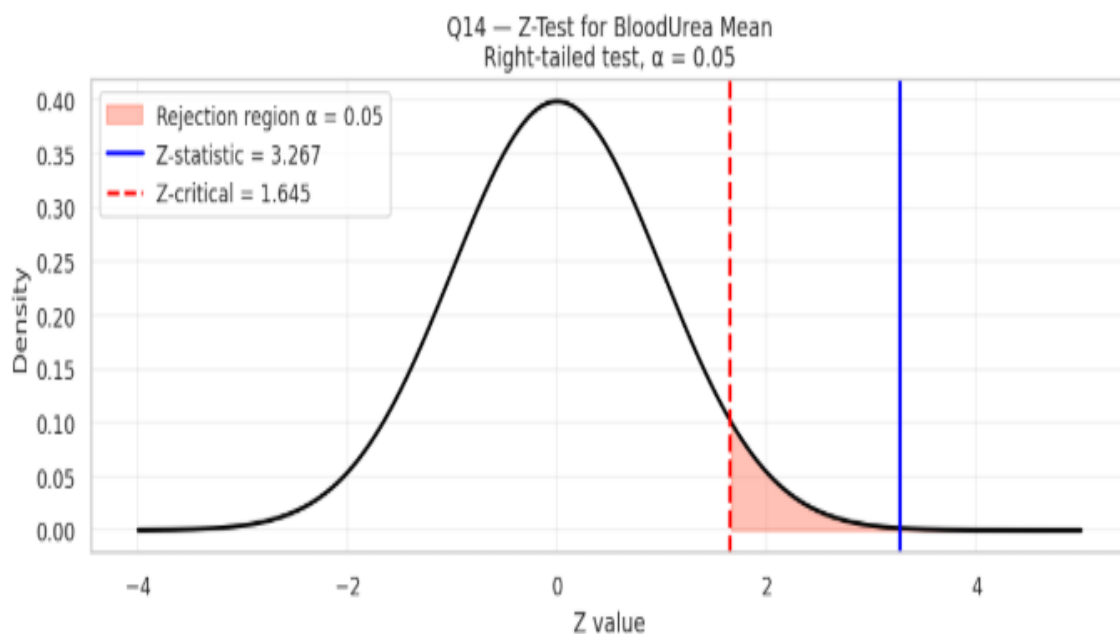
P-value = 0.000544

--- Decision ---

$Z = 3.2667 > Z_{\text{critical}} = 1.6449$

Decision: Reject H_0

Conclusion: There is sufficient statistical evidence at the 5% significance level that the average BloodUrea level is greater than 50 mg/dL.



--- Healthcare Interpretation ---

The test result shows that the sample mean BloodUrea level of 57 mg/dL is significantly greater than the claimed mean of 50 mg/dL.

Elevated BloodUrea may indicate reduced kidney filtration ability, which is commonly associated with Chronic Kidney Disease.

This result supports the need for clinical monitoring and further kidney function assessment among patients with high BloodUrea levels.

=====

QUESTION 15 – Hypothesis Test: Hemoglobin < 12.5 g/dL

=====

$H_0: \mu = 12.5$ g/dL

$H_1: \mu < 12.5$ g/dL

Significance level: $\alpha = 0.01$

Test type: One-tailed left-tailed t-test

Given: $n = 64$, $\bar{X} = 11.4$, $s = 1.8$

Degrees of freedom: $df = 63$

Standard Error: $SE = s/\sqrt{n} = 0.2250$

Test Statistic: $t = (\bar{X} - \mu_0) / SE = (11.4 - 12.5) / 0.2250 = -4.8889$

Critical Value: $t_{\alpha} = -2.3870$

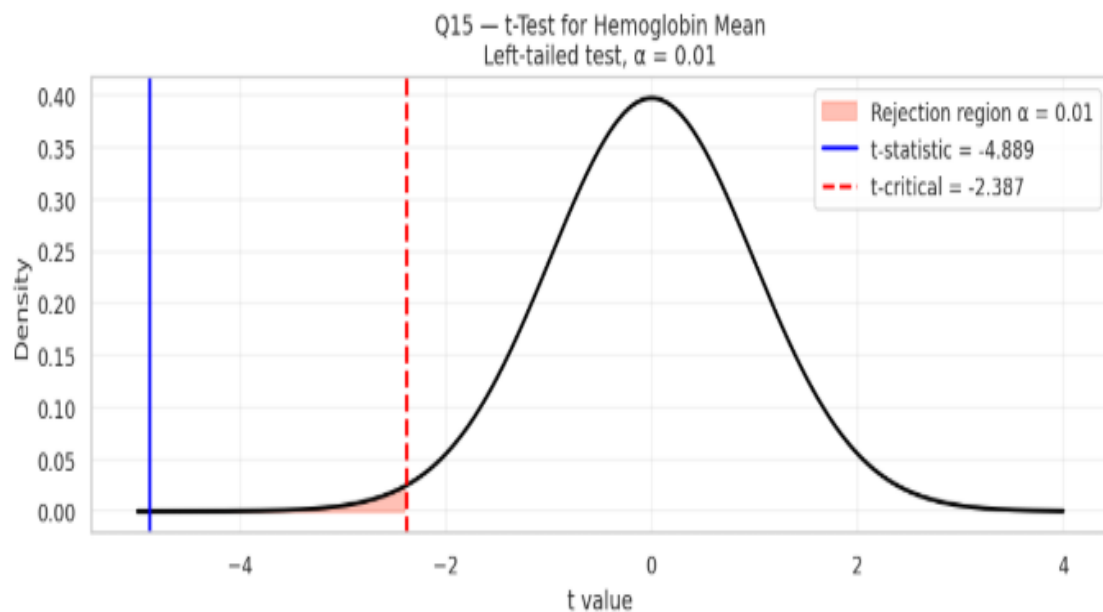
P-value = 0.000004

--- Decision ---

$t = -4.8889 < t_{\text{critical}} = -2.3870$

Decision: Reject H_0

Conclusion: There is sufficient statistical evidence at the 1% significance level that the average Hemoglobin level is less than 12.5 g/dL.



--- Healthcare Interpretation ---

The test result shows that the sample mean Hemoglobin level of 11.4 g/dL is significantly lower than the claimed mean of 12.5 g/dL.

Lower Hemoglobin levels may indicate anemia, which is commonly observed in Chronic Kidney Disease patients.

This occurs because damaged kidneys may produce less erythropoietin, reducing red blood cell production and lowering Hemoglobin levels.

=====

QUESTION 16 – Correlation Analysis

=====

Pearson Correlation Results:

	Pair	Valid N	r	p-value	Significance	Strength	Direction
0	SerumCreatinine vs BloodUrea	380	0.5864	0.000000	***	Moderate	Positive
1	BloodPressure vs SerumCreatinine	373	0.1462	0.004658	**	Weak	Positive
2	Hemoglobin vs CKD	348	-0.7689	0.000000	***	Strong	Negative
3	Potassium vs Sodium	312	0.0979	0.084306	NS	Weak	Positive

Significance Key:

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS = not significant

--- Healthcare Interpretation ---

SerumCreatinine vs BloodUrea: $r = 0.5864$, Moderate Positive correlation.

Both are kidney function markers; a positive correlation suggests that creatinine and urea tend to increase together when kidney filtration worsens.

BloodPressure vs SerumCreatinine: $r = 0.1462$, Weak Positive correlation.

A positive relationship may suggest that higher blood pressure is associated with reduced kidney function.

Hemoglobin vs CKD: $r = -0.7689$, Strong Negative correlation.

A negative correlation indicates that CKD patients tend to have lower Hemoglobin levels, which is consistent with anemia in CKD.

Potassium vs Sodium: $r = 0.0979$, Weak Positive correlation.

This relationship reflects electrolyte balance, which can be affected when kidney regulation is impaired.

=====

QUESTION 17 – Linear Regression: BloodPressure → SerumCreatinine

=====

Regression Equation: SerumCreatinine = -1.5727 + 0.0610 × BloodPressure

Slope β_1 = 0.060984

Intercept β_0 = -1.5727

R² value = 0.0214 (2.14% variance explained)

Pearson r = 0.1462

p-value = 0.004658

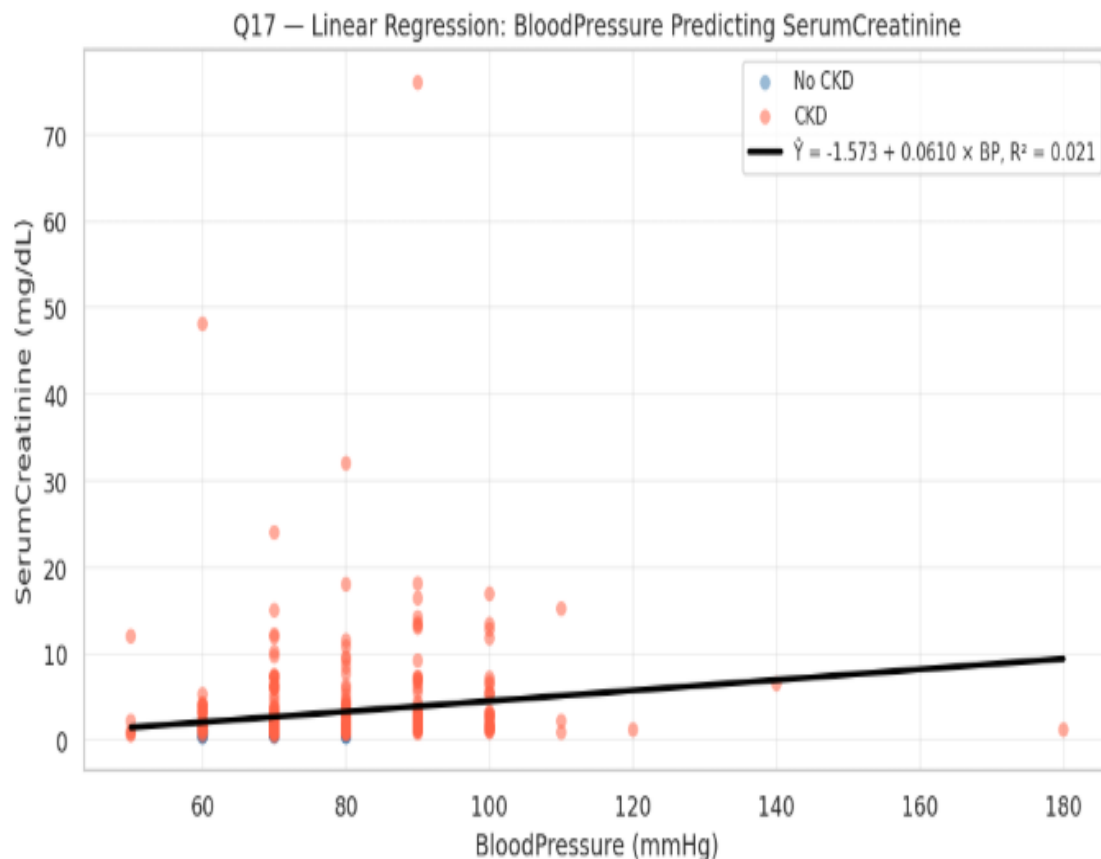
Sample Predictions:

BloodPressure = 70 mmHg → Predicted SerumCreatinine = 2.6962 mg/dL

BloodPressure = 80 mmHg → Predicted SerumCreatinine = 3.3060 mg/dL

BloodPressure = 90 mmHg → Predicted SerumCreatinine = 3.9158 mg/dL

BloodPressure = 100 mmHg → Predicted SerumCreatinine = 4.5257 mg/dL



--- Healthcare Interpretation ---

The slope value of 0.0610 means that for every 1 mmHg increase in BloodPressure, the predicted SerumCreatinine level changes by 0.0610 mg/dL.

The R² value of 0.0214 indicates that BloodPressure explains about 2.14% of the variation in SerumCreatinine.

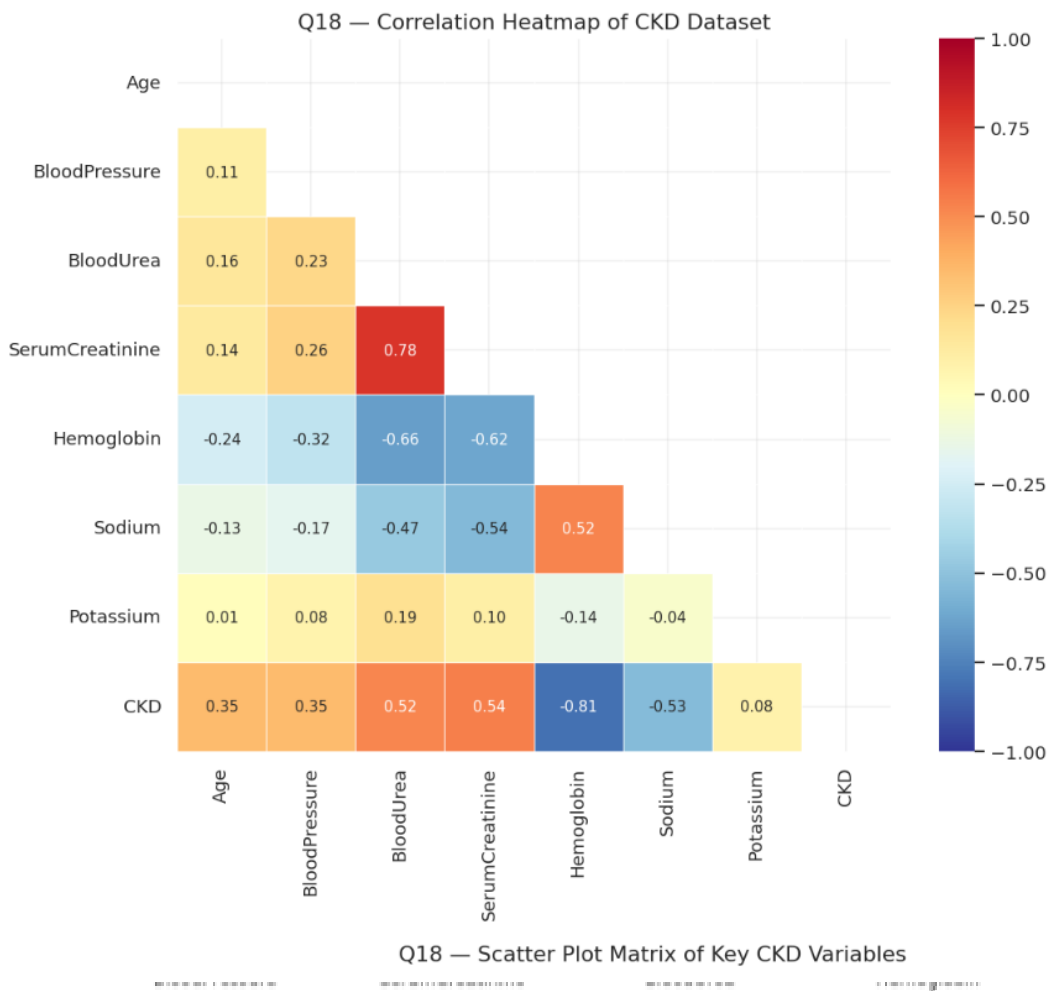
A positive relationship suggests that higher blood pressure may be associated with poorer kidney function, although BloodPressure alone has limited predictive power.

This supports the healthcare importance of blood pressure control in CKD prevention and management.

=====

QUESTION 18 – Correlation Heatmap & Scatter Plot Matrix

=====



--- Healthcare Interpretation ---

The correlation heatmap shows the strength and direction of relationships among important CKD-related variables.

SerumCreatinine and BloodUrea are expected to show a positive relationship because both are kidney function markers that rise when filtration worsens.

Hemoglobin is expected to show a negative relationship with CKD because CKD patients often experience anemia.

BloodPressure may show a positive relationship with CKD-related markers, supporting the role of hypertension as a risk factor for kidney disease.

The scatter plot matrix helps visually compare CKD and non-CKD patients and identify major risk factor patterns in the dataset.

=====

QUESTION 19 – CKD vs Non-CKD: Independent Samples t-Test

=====

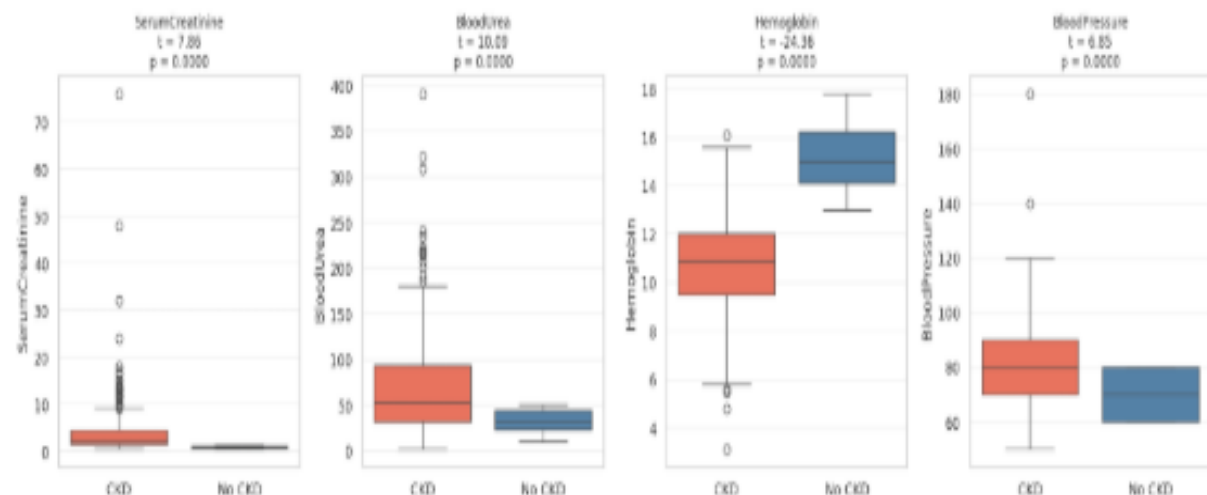
CKD patients : 250

Non-CKD patients : 150

Independent Samples t-Test Results:

	Variable	CKD Mean	Non-CKD Mean	t-statistic	p-value	Significant (p<0.05)
0	SerumCreatinine	4.415	0.869	7.8621	0.0	YES
1	BloodUrea	72.389	32.799	10.0907	0.0	YES
2	Hemoglobin	10.648	15.188	-24.3565	0.0	YES
3	BloodPressure	79.625	71.351	6.8469	0.0	YES

Q19 – Comparison of CKD and Non-CKD Patients



--- Healthcare Interpretation ---

SerumCreatinine: CKD mean is higher by 3.546. Statistically significant: YES.

BloodUrea: CKD mean is higher by 39.590. Statistically significant: YES.

Hemoglobin: CKD mean is lower by 4.540. Statistically significant: YES.

BloodPressure: CKD mean is higher by 8.274. Statistically significant: YES.

The independent samples t-test evaluates whether CKD and non-CKD patients differ significantly in important clinical measurements.

Significant differences suggest that the variable may be useful for distinguishing CKD patients from healthy individuals.

=====

QUESTION 20 – Complete Evidence-Based CKD Conclusion

=====

EVIDENCE-BASED CONCLUSION: CKD STATISTICAL ANALYSIS

1. Descriptive Analysis

CKD patients showed higher average SerumCreatinine and BloodUrea levels than non-CKD patients. These findings indicate reduced kidney filtration ability. CKD patients also showed lower Hemoglobin levels, suggesting anemia, and higher BloodPressure, supporting the relationship between hypertension and kidney disease.

2. Probability and Bayes Theorem

Conditional probability and Bayes theorem showed that patients with elevated SerumCreatinine have a much higher probability of having CKD. This supports SerumCreatinine as an important clinical screening indicator.

3. Distribution Analysis

The binomial distribution helped estimate CKD occurrence in a fixed group of patients. The normal distribution model was used to classify SerumCreatinine levels into normal, moderate-risk, and high-risk ranges. The uniform distribution model described patient age probability, while the exponential distribution modeled waiting time between CKD patient admissions.

4. Central Limit Theorem

The Central Limit Theorem showed that the sampling distribution of mean BloodUrea can be approximated by a normal distribution for a sufficiently large sample. This helps healthcare researchers make reliable conclusions about population-level kidney function using sample data.

5. Confidence Interval

The 95% confidence interval for mean SerumCreatinine was above the normal reference limit. This provides statistical evidence that CKD patients have clinically elevated SerumCreatinine levels.

6. Hypothesis Testing

The BloodUrea hypothesis test showed that the average BloodUrea level was significantly greater than 50 mg/dL. The Hemoglobin hypothesis test showed that the average Hemoglobin level was significantly less than 12.5 g/dL. These results support the presence of kidney dysfunction and anemia among CKD patients.

7. Correlation and Regression

Correlation analysis showed relationships among important CKD-related variables. SerumCreatinine and BloodUrea were positively related, while Hemoglobin was negatively associated with CKD status. Linear regression showed that BloodPressure has a positive relationship with SerumCreatinine, although BloodPressure alone may not fully explain kidney function variation.

8. Major CKD Risk Factors Identified

The major risk factors and indicators observed in this analysis were elevated SerumCreatinine, elevated BloodUrea, low Hemoglobin, high BloodPressure, electrolyte imbalance, hypertension, and diabetes mellitus.

9. Preventive Recommendations

Patients with hypertension or diabetes should regularly monitor SerumCreatinine, BloodUrea, BloodPressure, and Hemoglobin. Early screening, blood pressure control, diabetes management, anemia monitoring, and lifestyle modification can help reduce CKD progression risk.

10. Final Statistical Conclusion

Overall, the statistical analyses consistently show clear differences between CKD and non-CKD patients. SerumCreatinine, BloodUrea, Hemoglobin, and BloodPressure are important variables for understanding CKD risk. Combining descriptive statistics, probability, Bayes theorem, distribution, confidence intervals, hypothesis testing, correlation, regression, and visualization provides strong evidence that CKD occurrence can be statistically investigated and meaningfully predicted using healthcare data.